Paleomagnetic and Geochemical Constraints on the Timing and Duration of the CAMP Activity in Northeastern Brazil

M. Ernesto[1], G. Bellieni[2], E.M. Piccirillo[3], L.S. Marques[1], A. de Min[3], I.G. Pacca[1], G. Martins[4], and J.W.P. Macedo[5]

The northeastern border of the Maranhão basin (NE Brazil) is dominated by low-Ti tholeiites which show strong geochemical affinity with the low-Ti Early Jurassic Mosquito Formation from the western side of the basin. The low-Ti tholeiitic flows (Early Jurassic) from Lavras basin, and the low-Ti dikes outcropping in the Ceará State are also comparable to the Mosquito tholeiites. Compositional similarity also exists between the Early Cretaceous Sardinha intrusives (eastern Maranhão basin), the high-Ti dikes from the Ceará State, and the high-Ti rocks of the Rio Ceará-Mirim dike swarm (mainly subswarms I and III) easterwards. The high-Ti and low-Ti rocks show paleomagnetic remanences that allow the discrimination of the two groups. However, the characteristic remanent magnetization from the Rio Ceará-Mirim subswarm II differs significantly from the other two groups; these rocks can be also distinguished from the other high-Ti rocks by slight different chemical characteristics. On paleomagnetic basis, three age groups were identified: Early Cretaceous (Sardinha Formation, high-Ti Ceará dikes, and Ceará-Mirim subswarms I, III, and V), Early Jurassic (Mosquito Formation, low-Ti Ceará dikes, Lavras basin flow, and Ceará-Mirim subswarm IV), and Late Triassic (Ceará-Mirim subswarm II), and corresponding paleomagnetic poles were calculated. The Early Jurassic pole fits well with other CAMP poles from South America, extending easterwards the area in northern South America affected by that magmatic event. The Late Triassic age might represent a magmatic manifestation preceding by about 20-30 My the CAMP activity in NE Brazil.

[1]Departamento de Geofísica, Universidade de São Paulo, São Paulo, Brazil.

[2]Dipartimento di Mineralogia e Petrologia, Universita di Padova, Padova, Italy.

[3]Dipartimento di Scienze della Terra, Universita di Trieste, Trieste, Italy.

[4]Departamento de Geologia, Univerisidade Federal do Ceará, Fortaleza, Brazil.

[5]Departamento de Física, Univerisidade Federal do Rio Grande do Norte, Natal, Brazil.

The Central Atlantic Magmatic Province:
Insights from Fragments of Pangea
Geophysical Monograph 136
Copyright 2003 by the American Geophysical Union
10.1029/136GM07

INTRODUCTION

Northeastern Brazil (Fig. 1) was affected by recurrent tholeiitic magmatic activity [e.g. *Sial*, 1976; *Almeida et al.*, 1988; *Bellieni et al.*, 1990, and 1992] since at least Early Jurassic to Late Cretaceous. Basalt flows of the Early Jurassic Mosquito Formation [*Bellieni et al.*, 1990; *Fodor et al.*, 1990; *Baksi and Archibald*, 1997; and *Marzoli et al.*, 1999] are found in the western side of the Maranhão basin whereas in the eastern side, sills, dikes and scarce basaltic flows of the Early Cretaceous Sardinha Formation are present. Eastwards the Ceará-Mirim dike swarm [*Sial*, 1976; *Gomes et al.*, 1981; *Martins*, 1991; *Bellieni et al.*, 1992; *Oliveira*, 1992] trends east-west for more than 300 km. *Sial* [1975] subdivided the Ceará-Mirim dike swarm into six subparallel lineaments or subswarms of which five

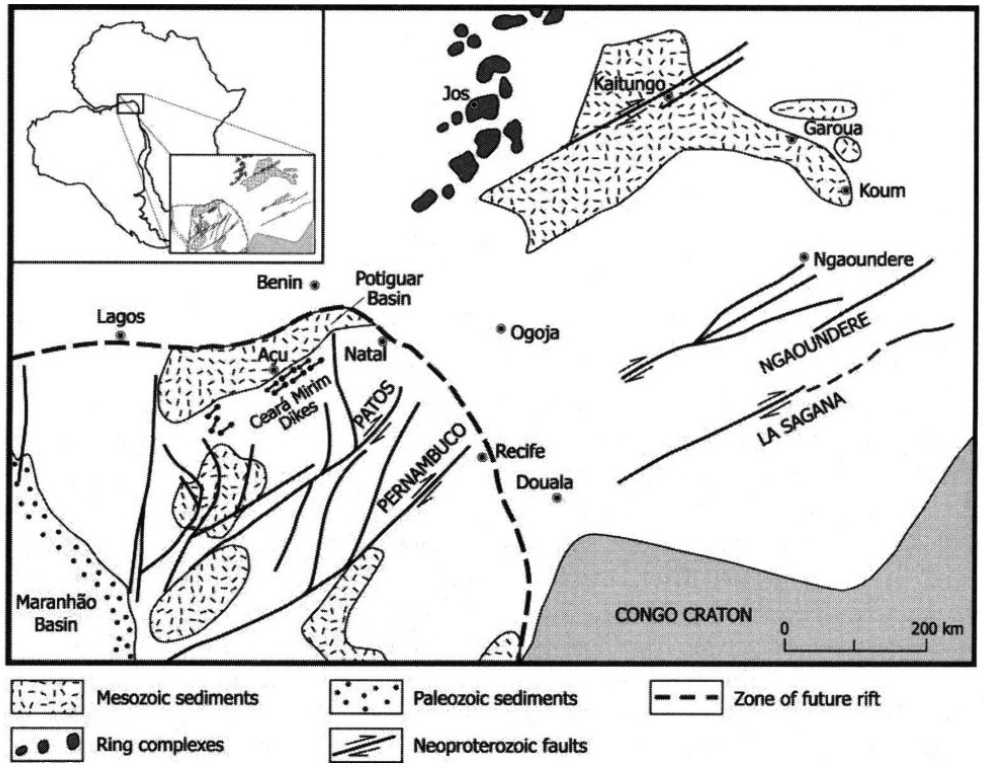


Figure 1. Schematic map of Northeastern Brazil showing the location of the Mosquito and Sardinha Formations (Maranhão basin), the Ceará-Mirim dike swarm, the Lavras da Mangabeira flows, and other magmatic occurrences in the Ceará and Piauí States. Full triangles represent the Tertiary volcanic necks and plugs. Sampling sites are also indicated. Dashed line represents the eastern limit of the Maranhão basin.

were also discriminated by paleomagnetic data [*Martins*, 1991; *Bellieni et al.*, 1992], and are shown in Fig. 2. Some isolated dikes trending E-W to NE-SW are also found in the Ceará State, between the Maranhão basin and the area occupied by the Ceará-Mirim dike swarm, forming a practically continuous magmatic area along Northeastern Brazil. The easternmost Early Jurassic rocks are represented by a few basalt flows [*Marzoli et al.*, 1999; and *De Min et al.*, this volume] located in the Lavras sedimentary basin (Fig 1).

In spite of the large amount of K-Ar, and some $^{40}Ar/^{39}Ar$ data for the Mesozoic magmatism in Northeastern Brazil, the age of the Ceará-Mirim dikes is still debatable. *Bellieni et al.* [1992] concluded that the distribution of the K-Ar and fission track ages delineated two maxima, 175-160 Ma and 145-125 Ma, indicating the existence of both Middle Jurassic and Early Cretaceous dikes in the Ceará-Mirim swarm. This was previously suggested by *Horn et al.* [1988] based on a minimum K-Ar age of 164±5 Ma [Middle Jurassic) calculated for two dikes belonging to subswarm II. However, K-Ar ages as old as 214±6 Ma and 228±3 Ma [*Bellieni et al.*, 1992] for dikes of subswarm II, and an age of 245±13 Ma [*Gomes et al.*, 1981] were discarded as anomalous data.

In a pre-drift reconstruction (Fig. 3) the area occupied by the Ceará-Mirim swarm lies opposite to the Benue Trough (Nigeria) where two Mesozoic periods of magmatic activity, 147-106 Ma, and 97-81 Ma have been recognized by *Maluski et al.* [1995]. However, northwards in the Jos plateau the alkaline ring complexes vary in age from 213 to 141 Ma [*Maluski et al.*, 1995, and references therein].

Paleomagnetic results for the Ceará-Mirim dikes, and Maranhão Basin magmatism were first reported by *Guerreiro and Schult* [1983] and *Schult and Guerreiro* [1979], respectively. However, only the Ceará-Mirim dikes cropping out east of Augusto Severo (Fig. 2), corresponding to subswarms I, II and III, were investigated, and an unique paleomagnetic pole was calculated. *Bücker et al.* [1986] studied the magnetic mineralogy of these dikes and improved the existing paleomagnetic results, although the recalculated pole did not change significantly. *Horn et al.* [1988] obtained K-Ar ages for the Ceará-Mirim dikes in the range 167 to 130 Ma and concluded that two different poles should be calculated: an Early Cretaceous pole based

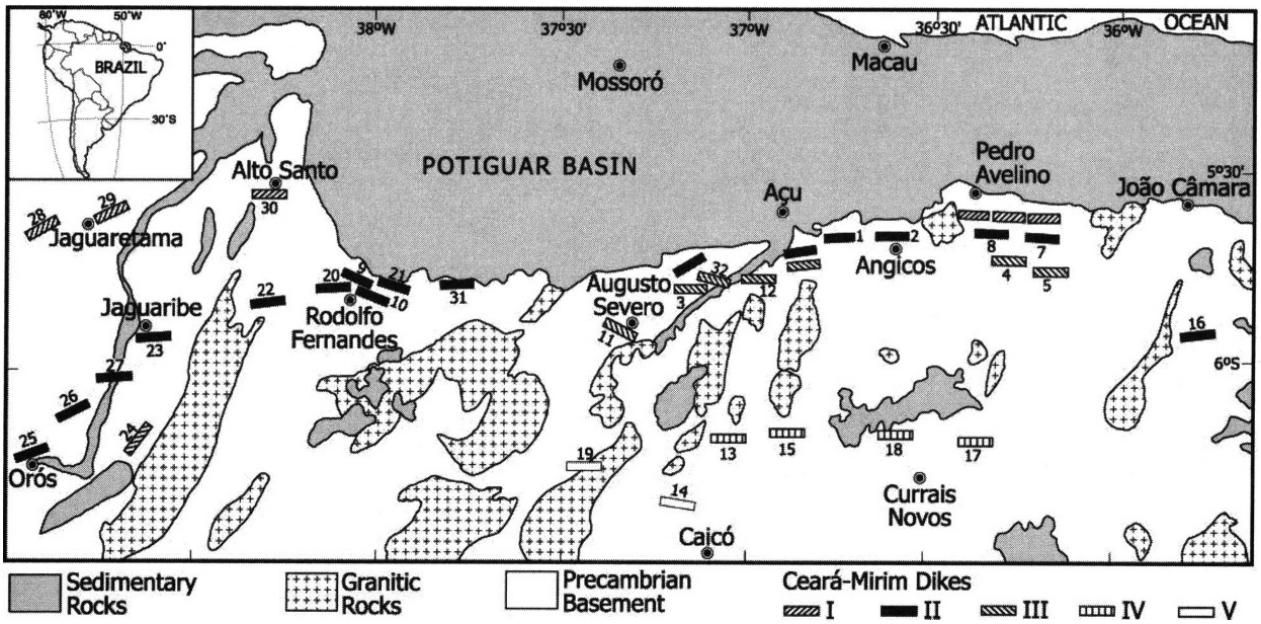


Figure 2. Sampling sites of the Ceará-Mirim dike swarm distinguishing dikes belonging to subswarms I to V, according to *Sial* [1975], and the magnetic characteristics (this paper).

on five sites from subswarms I and III, and a Jurassic pole based on five sites from subswarm II. The same subdivision was presented in *Martins* [1991] and *Bellieni et al.* [1992] for a higher number of sites.

Geochemical studies of both Early Jurassic and Early Cretaceous tholeiites of the Maranhão Basin [*Bellieni et al.*, 1990] evidenced that these two magmatic episodes can be distinguished by their low-Ti (Mosquito Formation) and high-Ti (Sardinha Formation) content, respectively. The same behavior is also observed for incompatible trace elements (e.g. Ba, La, Ce, Y, Zr, Sr), that also present relatively low and high concentrations in the Mosquito and Sardinha rocks, respectively. The Ceará-Mirim dike swarm has also been investigated by *Bellieni et al.* [1992]. Both Jurassic and Early Cretaceous dikes have high-Ti and incompatible trace element (ITE) concentrations, with the Jurassic ones showing slightly higher TiO_2 and ITE. Rare tholeiitic dikes low in TiO_2 and ITE (subswarm IV) are also present. *Bellieni et al.* [1992] concluded that the Ceará-Mirim tholeiitic dikes show close mineralogical, chemical and isotopic similarities with the Early Cretaceous high-Ti tholeiites from the Maranhão basin.

This paper presents new paleomagnetic data from the two age groups of the Maranhão basin, and extends the Ceará-Mirim sampling area including sites from subswarms IV and V (Fig. 2). Some isolated dikes outcropping to the west of the Ceará-Mirim swarm in the Ceará State, and two Jurassic flows from the Lavras basin were also studied. New geochemical data from the Maranhão basin and Ceará dikes are also presented. The analyzed samples come from the same paleomagnetic sites (Fig. 1). Comparisons of this new data set with the preexisting results from the same areas, along with paleomagnetic information, will be used in the investigation of the age, and spatial distribution of the different magmatic episodes in Northeastern Brazil.

GEOLOGICAL ASPECTS

The Ceará-Mirim dikes subvertically cross-cut the Precambrian crystalline basement to the south of the Mesozoic Potiguar sedimentary basin, in a general east-west direction. However, to the west they progressively change towards NE-SW (Fig. 2). Thickness generally ranges from 5 to 30m although it may reach about 100m [*Rolff*, 1965 and *Sial*, 1975]. According to *Bellieni et al.* [1992] the Ceará-Mirim magmatism is essentially represented by high-Ti ($TiO_2 > 2$ wt%) evolved tholeiites (andesi-basalts; mg number: 0.48-0.30) excepting for subswarm V which is constituted by alkaline rocks (tephrite and hawaiite). The Ceará-Mirim tholeiitic dikes show fine- to medium-grained size textures and mineral assemblages commonly consisting of plagioclase, augite, pigeonite, Ti-magnetite, apatite and ilmenite. In the tholeiitic dikes olivine is scarce, and the small crystals are completely altered. It is worth noting that for similar MgO content, the dikes belonging to sub-

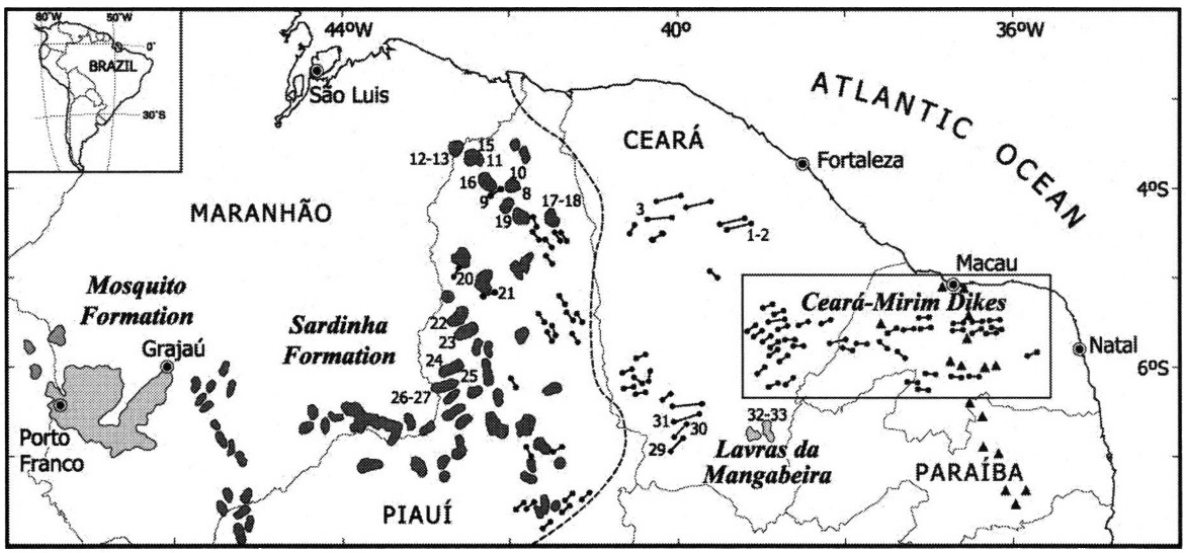


Figure 3. The Northeastern Brazil and the corresponding Northwestern African sector in a pre-drift reconstruction. Simplified from *Popoff* [1988].

swarm II show slightly higher TiO_2 and lower P_2O_5, K_2O, Y, Zr, La, Ce contents relative to the subswarms I and III.

The Paleozoic Maranhão sedimentary basin (Fig. 1) is located in Northern Brazil and covers about 600,000 km². During the Mesozoic it has been affected by a magmatic activity resulting in the emplacement of flows, dikes and sills. The Jurassic flows of Mosquito Formation appear restricted to the western side of the basin, whereas the Early Cretaceous intrusives of the Sardinha Formation (mainly dikes and sills) dominate towards the eastern border. The dikes mainly trend N50-70E and N20-40W [*Rezende*, 1964; this work) and thickness ranges from 2 to 60m. For the Sardinha Formation K-Ar ages [*Sial*, 1976] are close to 158 Ma, while $^{40}Ar/^{39}Ar$ data [*Baksi and Archibald*, 1997] vary from 129 to 124 Ma. However, *Nascimento et al.* [1981] reported a K-Ar age of 171±8 Ma in the northeastern area, which is in accordance with the 200-154 Ma [*Bellieni et al.*, 1990; *Fodor et al.*, 1990] interval defined by K-Ar ages for the Jurassic Mosquito Formation. A narrower interval of 200-190 Ma [*Baksi and Archibald*, 1997; *Marzoli et al.*, 1999] is defined by $^{40}Ar/^{39}Ar$ data for the Mosquito rocks.

The Maranhão magmatism is generally characterized by evolved basalts with mg number usually lower than 0.6. The Mosquito Formation is mostly represented by tholeiitic basalts with slightly porphyritic texture and mineral assemblages made up of augite, plagioclase, pigeonite and Ti-magnetite. Olivine, commonly confined in the groundmass, is scarce and completely altered. The Mosquito rocks have low concentrations of TiO_2, P_2O_5, K_2O, FeO_t and ITE. The high-Ti Sardinha intrusive rocks have fine- to medium-grained size and mineral assemblages consisting of augite, plagioclase, pigeonite, Ti-magnetite and scarce ilmenite, completely altered olivine and apatite. They are more evolved than the Mosquito volcanics, and mainly correspond to andesi-basalts [*Bellieni et al., 1990*].

The Lavras basin, near the locality of Lavras da Mangabeira, is a small Mesozoic sedimentary basin covering an area of about 70 km², and consisting of up to 80 m of sediments. Concordant vesicular basaltic flows occur at the base of the sedimentary column showing thickness of 1 to 8 m. These flows show slightly porphyritic to aphyric textures and mineral assemblages made up of augite, plagioclase, pigeonite, Ti-magnetite and scarce completely altered olivine. These basalts contain zeolite- and carbonate-bearing vesicules (L.O.I.=3.74 - 4.29 wt%) and are characterized by variable degree of alteration. Their tholeiitic nature is given by coexisting augite and pigeonite, while the elements less influenced by alteration (e.g. Ti, Zr, Y) indicate that they are of low-Ti type tholeiites, as the Mosquito ones. K-Ar data [*Priem*, 1978] for these basalts gave a mean age of 175±4 Ma, whereas one $^{40}Ar/^{39}Ar$ age [*Marzoli et al.*, 1999] is of 198.4±1.4 Ma.

During the mid-Tertiary (~29-36 Ma) alkaline magmatism occurred [*Sial et al.*, 1981; *Almeida et al.*, 1988] affecting the Ceará and Rio Grande do Norte States; necks and plugs (the Macau magmatism) cross-cut the Ceará-Mirim dike swarm (Fig. 1) in a nearly N-S direction.

PALEOMAGNETISM

Eighty-nine hand samples were taken from 32 sampling sites (dikes) of the Ceará-Mirim dike swarm, and 6 to 9 cylinders were cut in the field from 28 sites in the Maranhão basin and Ceará State. Sample orientation was performed by both magnetic and sun compasses whenever possible. One cylinder (2.5cm in diameter and 2.2cm long) of each sample was submitted to detailed AF cleaning up to 100mT (200mT in a few cases) in a Molspin - MSA2 tumbling demagnetizer and a second cylinder underwent detailed thermal cleaning (up to 600 °C) in a Schonstedt - TSD1 shielded furnace. A Molspin - MS1 spinner magnetometer was used for the remanence measurements. In general, the most stable component of magnetization was isolated by applying fields of moderate intensity but in some cases relatively high fields were necessary to erase secondary magnetizations as is shown by the vector diagrams in Fig. 4. Thermal cleaning was in general as efficient as AF cleaning but samples were previously submitted to a demagnetizing 10 to 30 mT peak field. During thermal cleaning samples became completely demagnetized at temperatures ranging from 530°C to 600°C. Characteristic components of magnetization for each sample were identified by means of vector analyses (Figs. 4 and 5) using the least square fit of *Kirshvink* [1980]. Site mean characteristic direction was calculated by giving unit weight to specimens, and uncertainties were evaluated by *Fisher*'s [1953] statistics.

Paleomagnetic results for the Ceará-Mirim sites are displayed in Table 1, along with those from *Bücker et al.* [1986). Wulff plot of the site mean magnetization directions is on Fig. 6. Subswarms I to IV are clearly identified in this figure. Only one site from subswarm V (alkaline dikes) gave reliable result and plots close to subswarm I data. The other analyzed site of group V gave anomalous direction probably due to weathering. The magnetically defined subswarms correspond to the lineaments shown in Fig. 2. Secondary components were mainly identified in samples from site groups II, III and IV (Fig. 7), and will be discussed later.

Results for the Maranhão basin and Ceará State are listed in Table 2, and site mean remanence directions are on Fig. 6. Data from the Mosquito and Sardinha Formations [*Schult and Guerreiro*, 1979] are also included in Fig. 6. Site means are discriminated by their content of TiO_2 (see section 5) in the low- and high-Ti groups. The low-Ti rocks, including the Lavras basin samples, plot close to the existent Mosquito data, whereas the high-Ti group is undistinguishable from the Sardinha data. Therefore, the Jurassic sites can be distinguished from the Cretaceous sites by their geochemical characteristics as well as by their magnetic polarity, mainly normal and reversed for the Jurassic and Cretaceous, respectively (Fig. 6).

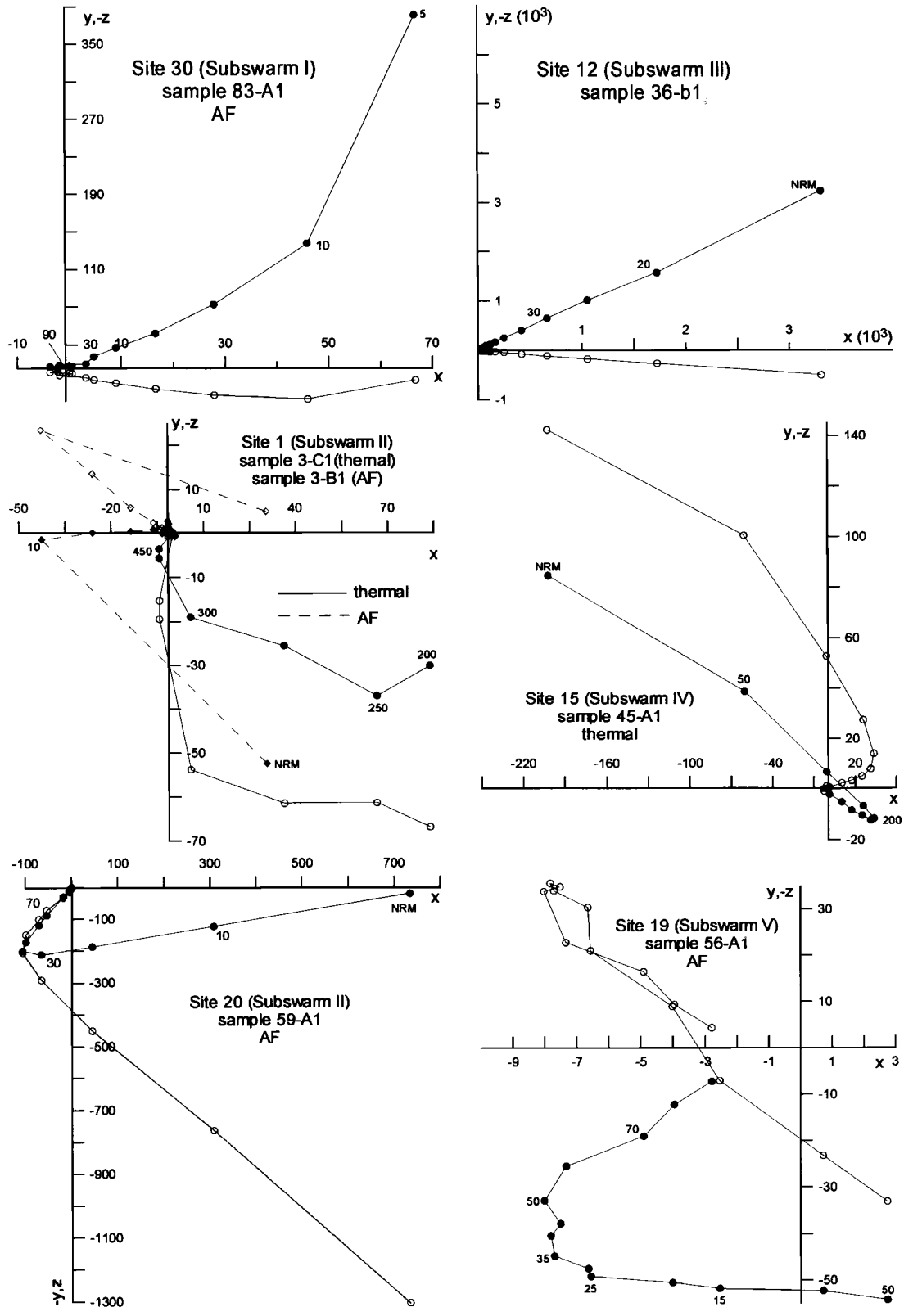
ROCKMAGNETISM

Rockmagnetic investigations by *Bücker et al.* [1986] in the Ceará-Mirim dikes revealed the presence of Ti-magnetites with low TiO_2 content and ilmenite exsolution lamellae as the result of high-temperature oxidation during slow cooling in the inner parts of the dikes. The origin of the magnetization was therefore partly attributed to a thermochemical rather than a thermal process owing to the low (500 to 600 C) temperatures calculated [*Buddington and Lindsley*, 1964] from Ti-magnetite and ilmenite compositions. Evidence of low-temperature oxidation, and hydrothermal alteration with the formation of maghemite was also found in the majority of samples.

The Fe-Ti oxides of samples belonging to the same dikes investigated here were analyzed by *Bellieni et al.* [1992]. They observed that Ti-magnetite is quite abundant (ca. 5-10 vol%) while ilmenite is comparatively scarce and is given by thin elongated laths. Ti-magnetite may be present in large crystals up to 4mm across. The compositions of homogeneous groundmass magnetite-ilmenite pairs yielded temperatures [*Buddington and Lindsley*, 1964] significantly higher than those reported previously: 1215 °C for a fine-grained sample from subswarm III, 740 °C for medium-grained samples from subswarm II and 721 °C for subswarm IV. These temperatures suggest a thermal origin for the remanence. Thermomagnetic analysis under fields of at most 400 mT ($3184 \times 10^{-2} Am^{-1}$) gave J_S/T curves of the same types as those reported by *Bücker et al.* [1986]. They are all indicative of Ti-magnetites highly oxidized with low TiO_2 content. About 60% of the investigated samples gave maximum-final to initial magnetization ratios (J_f/J_i) of about 0.70. Only 20% showed significant contents of maghemite.

Most of the Ceará-Mirim samples showed Curie temperatures in the range 536-559 C, except for samples from subswarm V with Curie temperatures as high as 596 C. The oxidation state (Fe^{3+}/Fe^{2+}) of the Ti-magnetites [*Bellieni et al.*, 1992] in the andesi-basalt rocks has an average of 0.48 for the early crystallized Ti-magnetites and 0.50 for the late crystallized ones. The lati-andesite group has lower ratios of 0.27 and 0.23 for the early and late crystallized minerals, respectively.

The low- and high-Ti samples from the Maranhão basin, and from other localities investigated in this paper, have Ti-magnetites as the main remanence carrier. Unblocking temperatures are in the range of 536-572 C, clustering


y,-z
Site 30 (Subswarm I)
sample 83-A1
AF
x
y,-z (10³)
Site 12 (Subswarm III)
sample 36-b1
NRM
x (10³)
y,-z
Site 1 (Subswarm II)
sample 3-C1(thermal)
sample 3-B1 (AF)
thermal
AF
NRM
y,-z
Site 15 (Subswarm IV)
sample 45-A1
thermal
NRM
Site 20 (Subswarm II)
sample 59-A1
AF
NRM
-y,z
y,-z
Site 19 (Subswarm V)
sample 56-A1
AF

around 560 C, indicating that the Ti-magnetites are highly oxidized at high temperatures. The Lavras basin samples yielded the lowest unblocking temperatures. Coercivity varies from 15 to 23 mT.

GEOCHEMISTRY

Major, minor and trace elements of 23 samples from the Maranhão basin and dikes from the Ceará State (Fig. 1; Table 3) were determined by using X-ray fluorescence, at the University of Trieste (Italy). Procedures are those described by *Bellieni et al.* [1983], and analytical precision is better than 3% for major and minor elements, whereas it is generally between 5 and 10% for trace elements. Selected samples from the Maranhão basin already investigated by *Bellieni et al.* [1990] were reanalyzed [*De Min et al.*, this volume] to provide a larger set of trace elements than presented before (Table 3). According to *De La Roche et al.* [1980], the investigated samples correspond to tholeiitic basalts, andesi-basalts and lati-andesites. Only two samples plot in the field of dacites (Fig. 8).

The basic rocks (SiO_2 < 54 wt%) have fine- to medium-grain size textures and mineral assemblages consisting of augite (Wo_{44-32}), plagioclase (An_{69-26}), pigeonite (Wo_{9-12}), Ti-magnetite (av. ulvöspinel 49%), apatite and occasionally ilmenite. Olivine is scarce, occurring as small completely altered crystals. The dacites present medium-grained size textures and are made up of plagioclase (An_{41-57}), augite (Wo_{31-41}), K-feldspar, Ti-magnetite, abundant quartz (more than 20 vol%) and apatite as accessory.

The investigated basic rocks are somewhat evolved, presenting low atomic $Mg/(Mg+Fe^{2+})$ ratios (mg number < 0.61). For similar MgO, the basic rocks may be divided into two groups, with very different geochemical characteristics (Fig. 9). The low-Ti group is characterized by low concentrations of TiO_2 (< 2 wt%), FeO_t, K_2O, P_2O_5 and ITE (e.g. light REE, Y, Zr, Nb and Sr), whereas the high-Ti (TiO_2 > 2%) group has higher abundance of such elements. Note that the low-Ti rocks (tholeiitic basalts and andesi-basalts) are lesser evolved (mg number: 0.47-0.61) than the high-Ti ones (andesi-basalts and lati-andesites) (mg number: 0.35-0.48). The high-Ti tholeiites have lower $SiO_2/(Na_2O+K_2O)$ values, straddling the basaltic trachyandesite field of the TAS diagram [*Le Bas et al.*, 1986; inset of Fig. 8].

The geochemical characteristics (Figs. 8 and 9), spatial distribution (Fig. 1) and paleomagnetic results (Fig. 6) make apparent that the analyzed high-Ti magmatism is very similar to that of the Sardinha Formation (Early Cretaceous). The new data strengthen the compositional similarity between the Sardinha intrusives and the high-Ti tholeiitic dikes from the Ceará-Mirim (Fig. 9), as previously shown by *Bellieni et al.* [1992]. Analogously, the low-Ti tholeiites presented in this paper (Figs. 8 and 9) show geochemical and paleomagnetic signatures very similar to the Early Jurassic basalt flows (Mosquito Formation) of the Maranhão basin [*Bellieni et al.*, 1990; *De Min et al.*, this volume]. Besides that, these low-Ti basalts are also very similar in composition (e.g. Ti, Zr, La, Ce, Y; Fig. 9) to the Early Jurassic low-Ti tholeiites from the Lavras basin [*De Min et al*, this volume].

The two analyzed dacites have important geochemical differences (Tab. 3; Fig. 8). The dacite from site 31 is significantly enriched in SiO_2, Na_2O, K_2O and incompatible trace elements relative to the dacite of site 17. These felsic magmas are believed derived from low-Ti tholeiitic melts through complex petrogenetic processes [*Novello*, 1995]. The paleomagnetic data indicate that dacites are related to the Jurassic tholeiitic magmatism, supporting the hypothesis of the involvement of low-Ti parental magmas in their genesis.

DISCUSSION AND CONCLUSIONS

Virtual geomagnetic poles (VGP) calculated for each site mean are displayed in Tables 1 and 2, and plotted in Fig. 10. VGPs from subswarms I and III are very coherent with the results from the Sardinha Formation, and the other high-Ti intrusive rocks. This comparison is also supported by the chemical characteristics of the corresponding rocks (Figs. 8 and 9), as discussed in the previous section. Although of different geochemical composition (alkaline rocks), subswarm V may be also incuded in the Lower Cretaceous paleomagnetic group. However, the high- and low-Ti subswarm IV seems to be an independent data set, comparable only with the low-Ti Jurassic data from the Lavras basin flow, and the Mosquito Formation (including also the other low-Ti dikes from the Maranhão basin). Therefore, it is suggested that subswarm IV might be of Early Jurassic age. VGPs from the low- and high-Ti dikes from the Ceará State are also in accordance with the Mosquito and Sardinha data, respectively. Therefore, two well characterized groups can be considered: the Early Cretaceous (Sardinha, Ceará-Mirim I, III and IV, and high-Ti

Figure 4. (Opposite.) Orthogonal vector plots for the Ceará-Mirim samples. Full symbols, horizontal plane; open symbols, vertical plane. AF demagnetization in mT, and thermal demagnetization in degrees Celsius.

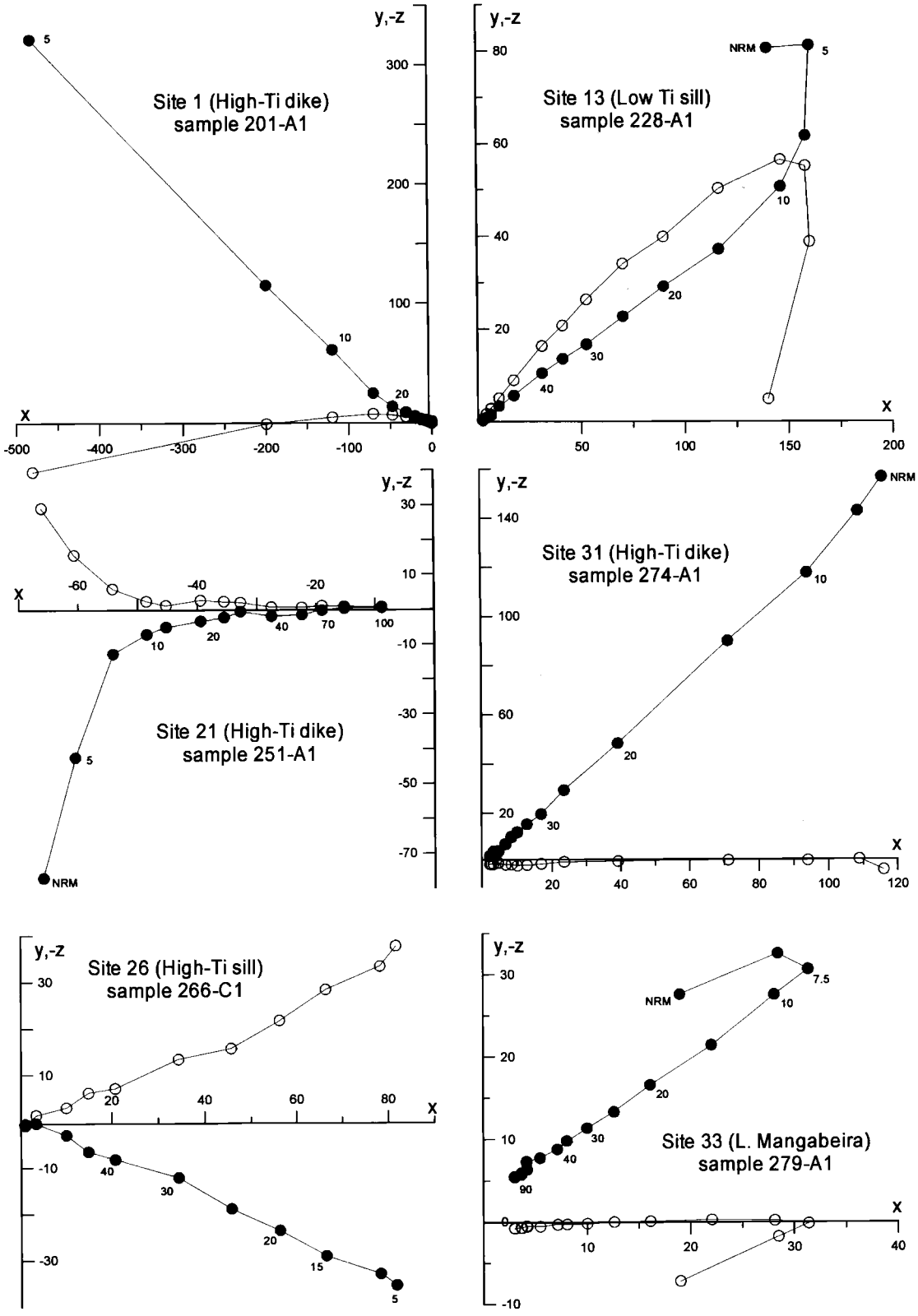

Site 1 (High-Ti dike)
sample 201-A1
y,-z
x
Site 13 (Low Ti sill)
sample 228-A1
NRM
Site 21 (High-Ti dike)
sample 251-A1
Site 31 (High-Ti dike)
sample 274-A1
Site 26 (High-Ti sill)
sample 266-C1
Site 33 (L. Mangabeira)
sample 279-A1

Table 1. Paleomagnetic results for the Ceará-Mirim dikes

Code	Site Long. (°W)	Site Lat. (°S)	N	Mean Magnetization Direction Dec. (°)	Inc. (°)	α_{95} (°)	k	Virtual Geomagnetic Pole Long. (°)	Lat. (°)	dp (°)	dm (°)
Subswarm I											
28	38°57'	5°39'	6	175.2	-9.9	2.7	607	116.7	-78.3	5.4	2.7
29	38°47'	5°37'	6	169.3	-3.9	9.6	49	86.2	-76.9	19.2	9.6
30	38°20'	5°37'	3	216.6	2.8	6.1	408	226.7	-53.2	12.2	6.1
44†			9	177.2	-5.3	2.6	387	125.3	-81.1	5.2	2.6
49†			7	178.2	-2.3	6.1	97	129.2	-82.8	12.2	6.1
Subswarm II											
1	36°46'	5°38'	4	198.4	28.4	10.5	77	261.1	-69.9	21.0	11.9
2	36°36'	5°38'	9	185.2	45.8	5.7	83	311.1	-67.8	11.4	8.2
6				random							
7	36°12'	5°37'		random							
8	36°21'	5°37'	9	184.2	54.3	5.2	98	316.6	-60.5	10.4	8.9
9	38°03'	5°46'	9	179.6	60.5	5.1	104	322.5	-54.3	10.2	10.4
10	38°03'	5°48'	5	225.8	43.8	6.1	158	261.6	-41.9	12.2	8.4
16	35°46'	5°54'	9	183.2	22.8	5.4	90	296.5	-83.2	10.8	5.9
20	38°05'	5°47'	9	196.5	51.3	6.7	61	293.4	-59.6	13.4	10.7
21	37°57'	5°46'	3	207.4	48.5	3.1	1569	277.8	-54.9	6.2	4.7
22	38°27'	5°50'	9	185.6	53.4	4.9	108	311.3	-61.4	9.8	8.2
23	38°47'	5°57'	8	194.6	42.7	4.9	126	286.5	-66.3	9.8	6.6
25	38°57'	6°15'	6	159.0	48.1	6.3	112	359.8	59.7	12.6	9.4
26	38°52'	6°10'	3	184.5	61.5	8.7	202	315.5	-53.3	17.4	18.2
27	38°50'	6°03'	6	70.3	-57.2	10.7	40	89.2	19.1	21.4	19.7
31	37°50'	5°48'	9	187.3	51.1	2.2	527	308.3	-63.1	4.4	3.5
43†			9	190.1	51.6	3.9	174	305.2	-61.7	6.1	5.3
47†			8	210.1	33.5	4.2	169	259.0	-57.9	7.8	4.8
50†			5	183.5	38.6	15.8	25	312.0	-73.5	28.3	18.8
B2†			6	191.9	40.1	3.2	436	290.0	-69.4	6.4	4.2
B4†			4	196.1	34.5	8.3	125	275.0	-69.5	16.6	10.1
Subswarm III											
3	37°29'	5°50'		random							
4	36°16'	5°42'	6	355.6	0.5	6.9	93	287.1	82.6	13.8	6.9
5	36°10'	5°44'	5	355.1	0.6	10.6	53	284.6	82.2	21.2	10.6
11	37°19'	5°53'	9	8.0	2.8	7.2	52	10.6	79.2	14.4	7.2
12	36°58'	5°46'	6	1.5	-0.6	4.2	247	338.4	84.3	8.4	4.2
24	38°40'	6°13'	6	359.1	-26.5	9.2	53	147.7	82.2	18.4	10.3
32	36°54'	5°46'		random							
51†			8	1.9	-11.1	5.8	91	51.0	88.1	11.5	5.9
46†			14	4.2	-3.3	4.1	91	10.5	84.1	8.2	4.1
B5†			7	0.9	-0.9	5.1	139	153.4	84.7	10.2	5.1
Subswarm IV											
13	37°00'	6°11'		random							
15	36°52'	6°10'	3/6	209.0	0.7	2.4	739	222.9	-60.5	4.8	2.4
17	36°21'	6°11'	3/9	203.2	-15.2	5.6	83	202.9	-63.0	11.2	5.8
18	36°35'	6°10'	3/8	196.6	-18.2	2.6	471	190.4	-67.3	5.2	2.7
Subswarm V											
14	37°10'	6°21'	3/8	331.3	32.4	7.6	55	273.5	52.9	15.2	9.0
19	37°24'	6°16'	2/5	182.7	12.6	2.7	808	235.1	-87.3	5.4	2.7

N, number of specimens for mean calculations; Dec. and Inc., Declination and Inclination; α_{95} and k, *Fisher*'s [1953] statistical parameters; Pol., polarity (N=normal, R=reversed); Long. and Lat., Longitude and Latitude; dp and dm, confidence oval of 95%. †Data from *Bücker et al.* [1986].

Figure 5. (Opposite.) Orthogonal vector plots for the Maranhão basin samples and Ceará dikes. Full symbols, horizontal plane; open symbols, vertical plane. AF demagnetization in mT, and thermal demagnetization in degrees Celsius.

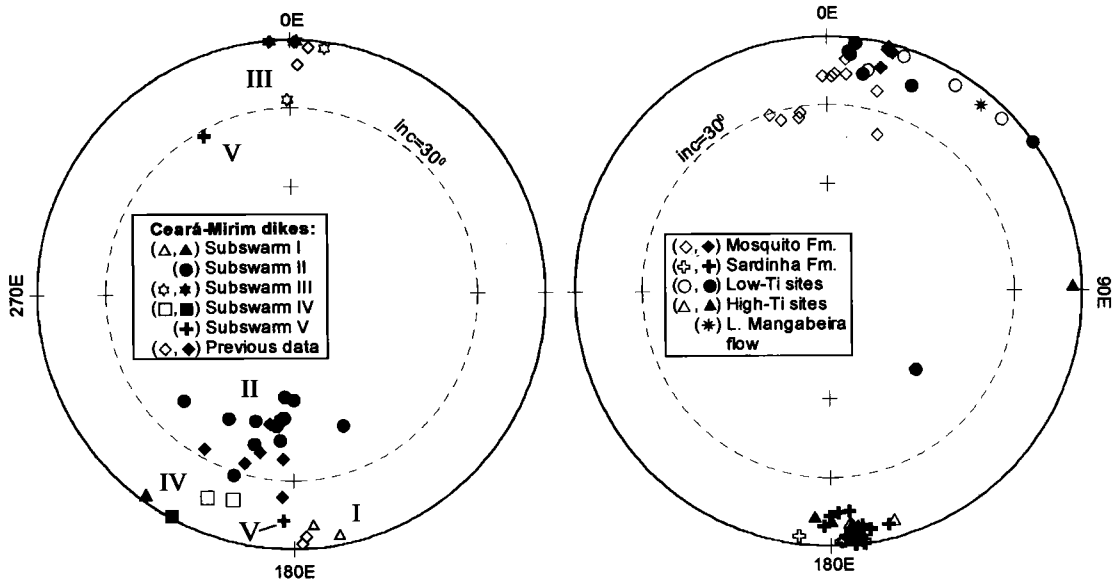


Figure 6. Site mean magnetization directions for a) Ceará-Mirim dikes, including data from *Bücker et al.* [1986], and b) Maranhão basin and Ceará dikes, and the Mosquito and Sardinha data from *Schult and Guerreiro* [1979]. Open and full symbols correspond to negative and positive inclinations, respectively.

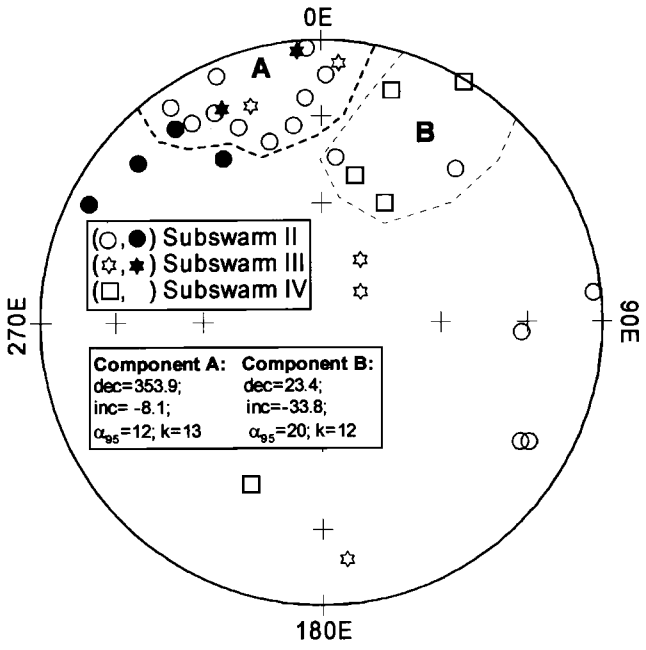


Figure 7. Secondary magnetization components identified in the Ceará-Mirim dikes. Open and full symbols correspond to negative and positive inclinations, respectively.

sites), and the Early Jurassic (Mosquito, Lavras basin, Ceará-Mirim IV, and low-Ti sites) groups. Corresponding paleomagnetic poles were calculated, and are referred as to Northeastern Brazilian Magmatism poles in Table 4.

Results from subswarm II are more striking in the sense that they do not match either the Early Cretaceous or the Early Jurassic data, although a Jurassic age is currently assigned in literature [*Bücker et al.*, 1986; *Horn et al.*, 1988; *Bellieni et al.*, 1992]. Therefore, an independent paleomagnetic pole was calculated based on this subswarm. Note that slight geochemical differences between subswarm II and the other Ceará-Mirim subswarms (Fig. 9) were already pointed out by *Bellieni et al.* [1992]. These differences are also valid in relation to the subswarm IV that fits well the Early Jurassic data.

Paleomagnetic poles calculated for the Early Cretaceous (NBM1), Early Jurassic (NBM2), and the Ceará Mirim subswarm II (CM2) are compared with the apparent polar wander path (APWP) for South America (Fig. 11). This APWP is well defined for the Early Cretaceous based on the robust Serra Geral [*Ernesto et al.*, 1999] and other time-equivalent poles from southeastern Brazil, Paraguay and Argentina (Table 4). The Early Cretaceous NBM1 pole plots in good agreement with the ~130 Ma segment of the polar path. The Jurassic pole calculated in this paper (NBM2) plots between the Late Jurassic poles from Argentina [*Rapalini et al.*, 1993, and references therein; *Vizán*, 1998], and the Early Jurassic (~200 Ma) from the Central Atlantic Magmatic Province - CAMP [*Marzoli et al.*, 1999]. Actually, NBM2 pole fits other Brazilian CAMP poles obtained from the Roraima and Cassiporé tholeiitic dikes [*Marzoli et al.*, 1999; *De Min et al.*, this volume; unpublished data) and the CAMP pole from the

Table 2. Paleomagnetic results from the Low- and High-Ti rocks, and Lavras da Mangabeira flows

	Sampling Sites				Mean Magnetization Direction						Virtual Geomagnetic Pole			
Code	Long. (°W)	Lat. (°S)	Trend	TiO_2	N	Dec. (°)	Inc. (°)	α_{95} (°)	k	Pol.	Long. (°)	Lat. (°)	dp (°)	dm (°)
High-Ti														
1	39°16'44"	4°21'57"	E-W	3.10	3	164.5	-7.9	6.7	339	R	78.9	-72.4	13.4	6.7
2	39°15'45"	4°21'41"	N80E	3.12	6	179.8	11.2	5.3	163	R	328.6	-88.5	10.5	5.4
21	42°20'06"	5°11'39"	N70E	3.41	1	188.9	- 4.1	4.0	121	R	187.3	-78.3	8.0	4.0
					2									
22	42°38'3"4	5°32'01"	sill	3.45	1	176.0	5.7	3.1	195	R	81.0	-85.2	6.2	3.1
					2									
23	42°35'26"	5°39'30"	sill	3.41	6	173.4	8.2	8.5	63	R	60.3	-83.2	16.9	8.6
24	42°45'11"	6°05'24"	sill	3.25	9	174.7	9.2	2.6	404	R	62.7	-84.5	5.2	2.6
25	42°43'28"	6°06'38"	sill	2.92	6	174.3	3.6	2.9	532	R	84.2	-82.8	5.8	2.9
26	42°49'39"	6°14'47"	sill	3.18	6	177.6	1.9	2.8	571	R	112.8	-84.2	5.6	2.8
27	42°50'34"	6°15'02"	sill	3.22	6	187.4	3.5	3.3	405	R	195.8	-81.1	6.6	3.3
Low-Ti														
3	40°08'53"	4°13'59"	N80E	1.19	3	9.3	-1.4	10.3	146	N	29.7	80.1	20.6	10.3
8	41°57'44"	4°03'30"	?	1.29	1	12.2	-16.8	2.4	338	N	69.4	77.1	4.7	2.5
					2									
9	42°08'02"	3°58'03"	N40W	1.63	3	5.2	6.3	5.9	431	N	353.9	81.2	11.8	5.9
10	42°10'56"	3°56'08"	flow	1.27	6	32.2	- 5.8	7.0	92	N	47.2	57.9	14.0	7.0
11	42°17'33"	3°45'52"	flow	1.08	6	9.5	15.6	7.0	94	N	356.6	74.9	13.8	7.2
12	42°37'44"	3°40'23"	sill?	1.58		ramdom								
13	42°35'12"	3°41'13"	sill?	1.85	9	18.3	- 3.7	6.9	57	N	42.3	71.6	13.8	6.9
14	42°32'24"	3°41'56"	sill?	1.19	3	10.6	-13.6	4.6	708	N	65.3	78.9	9.1	4.7
15*	42°26'56"	3°38'09"	sill?	0.95	2	22.6	14.8	10.7	544	N	20.9	64.8	21.2	11.0
16	42°13'58"	3°54'13"	sill?	1.35	6	6.7	2.2	6.4	109	N	9.1	81.3	12.8	6.4
17	41°28'32"	4°26'01"	sill	1.85	9	5.7	7.6	4.2	152	N	353.2	79.9	8.4	4.2
18*	41°28'29"	4°24'13"	sill?	1.44	6	132.4	56.7	5.8	133	R	4.3	-35.2	11.2	10.3
19	41°51'29"	4°23'39"	sill?	0.96	3	184.0	12.8	2.8	188	R	255.9	-85.5	5.6	2.9
									5					
20	42°36'56"	4°52'17"	N30E	1.59	6	45.7	- 4.4	8.5	62	N	45.8	44.3	17.0	8.5
29	40°00'40"	6°48'21"	N40E	1.99	3	183.5	26.6	11.5	116	R	294.9	81.9	23.0	12.8
30	39°56'29"	6°40'18"	N40E	2.21	5	ramdom								
31	39°50'15"	6°34'44"	N55E	1.51	3	54.4	0.3	5.7	470	N	45.3	35.3	11.4	5.7
Lavras da Mangabeira														
32	38°58'38"	6°48'04"	flow	1.06	6	ramdom								
33	38°57'25"	6°47'30"	flow	1.09	6	39.9	6.1	9.8	47	N	41.8	49.1	19.4	9.8

N, number of specimens for mean calculations; Dec. and Inc., declination and inclination; α_{95} and k, *Fisher*'s [1953] statistical parameters; Pol., polarity (N=normal, R=reversed); Long. and Lat, Longitude and Latitude; dp and dm, confidence oval of 95%; * Data not included in the means.

French Guyana dikes [*Nomade et al.*, 2000]. These poles tend to be slightly younger (possibly ~190 Ma) than the other CAMP group (Anari-Tapirapuã flows, and Amazonian sills; Table 4). This age interval (200-190) for the CAMP activity had already been predicted by *De Min et al.* [this volume].

The new CM2 pole obtained from the Ceará-Mirim subswarm II does not match the CAMP groups but is close to some of the available Triassic poles for South America, mainly those from the northern part of the continent (Table 4). Poles from the southern part (poles PV, IS, TE from Argentina; Fig. 11) form a distinct group closer to the Jurassic poles. This could be an indication of imprecise age definition of the rocks or even the presence of secondary magnetizations. However, if these poles do indicate a very slow movement of the South American plate during the Triassic-Jurassic interval, then the CM2 pole could be even older. The existence of K/Ar Triassic ages like 214±6 Ma (dike 2), 228±3 Ma (dike 16) obtained by *Bellieni et al.* [1992] for subswarm II, and the ~213 Ma age reported for the Jos plateau [*Maluski et al.*, 1995] in the contiguous portion of Africa (Fig. 3), point to a Triassic age for CM2

Table 3. Bulk rock compositions of low- and high-Ti rocks from the Maranhão basin and Ceará State.

Site	1	2	3	8	9	10	11	13	14	15	16	17	18	19	20	21	22
Sample	**8200**	**8201**	**8202**	**8207**	**8208**	**8209**	**8210**	**8212**	**8213**	**8214**	**8215**	**8216**	**8217**	**8218**	**8219**	**8220**	**8221**
SiO_2 (wt%)	53.87	53.73	52.82	52.82	54.13	52.60	52.75	54.03	52.42	52.78	53.24	59.91	53.44	52.83	51.89	53.70	53.79
TiO_2	3.10	3.12	1.19	1.07	1.56	1.04	1.08	1.85	1.19	0.95	1.35	1.85	1.44	0.96	1.59	3.41	3.45
Al_2O_3	13.67	13.26	15.49	14.80	13.71	14.63	15.62	12.52	15.11	15.22	13.91	11.91	14.72	15.39	14.60	14.01	13.50
FeO_t	12.81	12.68	10.26	9.79	12.36	9.73	9.76	13.21	10.29	9.22	11.40	13.44	11.47	9.87	11.95	12.26	12.60
MnO	0.19	0.19	0.17	0.17	0.19	0.16	0.16	0.20	0.17	0.16	0.19	0.21	0.19	0.17	0.21	0.19	0.19
MgO	3.36	3.73	5.93	7.53	5.42	7.68	6.67	5.68	6.49	7.26	6.58	1.51	5.56	7.07	5.60	3.29	3.46
CaO	6.65	7.47	10.68	11.00	9.17	11.46	10.82	8.81	11.06	11.53	10.28	5.43	9.32	10.74	10.92	7.47	6.93
Na_2O	3.07	2.76	2.50	2.14	2.43	2.06	2.40	2.52	2.36	2.20	2.34	3.14	2.73	2.24	2.39	2.70	2.98
K_2O	2.44	2.20	0.80	0.57	0.84	0.54	0.62	0.99	0.78	0.57	0.55	2.04	0.97	0.62	0.72	2.20	2.22
P_2O_5	0.84	0.86	0.16	0.11	0.19	0.10	0.11	0.20	0.13	0.10	0.16	0.56	0.17	0.11	0.13	0.77	0.88
Fe_2O_3	3.21	2.29	1.67	1.89	3.09	1.78	1.14	3.24	2.10	1.97	1.98	4.30	2.91	2.14	3.09	3.16	5.38
FeO	9.92	10.62	8.76	8.09	9.58	8.13	8.73	10.29	8.40	7.45	9.62	9.57	8.85	7.94	9.17	9.42	7.76
L.O.I.	2.02	1.99	0.80	1.31	1.60	1.03	1.08	1.71	1.17	1.02	1.47	1.89	1.44	1.29	2.58	1.87	1.76
mg number	0.35	0.37	0.54	0.61	0.47	0.61	0.58	0.47	0.56	0.61	0.54	0.19	0.50	0.59	0.49	0.35	0.36
Rock Type	LA	AB	TB	TB	AB	TB	TB	AB	TB	TB	TB	DAC	AB	TB	TB	AB	LA
Cr (ppm)	38	37	219	217	31	234	197	20	184	169	67	1	67	104	100	46	10
Ni	37	25	60	104	43	113	104	47	107	100	66	7	64	85	82	34	26
Ba	623	601	196	155	186	115	124	234	316	106	163	440	194	129	212	963	674
Rb	72	53	22	14	24	15	15	29	19	14	14	57	26	18	19	51	49
Sr	606	564	241	185	197	186	185	194	203	215	199	201	238	214	221	692	615
La	42	47	14	8	12	7	7	13	9	8	11	37	12	7	11	39	49
Ce	101	96	30	23	34	22	24	40	20	24	28	85	40	19	31	98	113
Nd	55	52	16	13	20	11	13	22	14	12	17	46	16	9	17	52	59
Zr	330	332	119	93	145	91	94	171	106	85	121	363	129	88	135	308	359
Y	50	51	28	24	34	24	25	38	28	34	30	76	32	26	33	46	51
Nb	27	28	10	6	9	6	6	11	7	6	8	22	9	7	10	27	30

Table 3. Continued

Site	23	24	25	26	27	31	SF	SF	SF	SF	SF	SF	SF	SF	SF	SF	SF
Sample	**8222**	**8223**	**8224**	**8225**	**8226**	**8230**	**5001**	**5002**	**5005**	**5006**	**5046**	**5047**	**5048**	**5049**	**5050**	**5051**	**5052**
SiO_2 (wt%)	52.61	54.32	52.05	52.99	52.84	61.74	52,92	53,03	52,20	52,33	53,97	52,03	51,36	55,92	53,59	52,36	51,07
TiO_2	3.41	3.25	2.92	3.18	3.22	1.51	3,05	3,40	3,34	3,24	3,33	4,01	4,09	2,41	3,31	3,52	3,84
Al_2O_3	13.10	13.10	13.83	13.01	12.95	14.81	13,77	14,62	14,47	14,72	13,87	14,63	14,30	13,18	14,15	13,07	14,80
FeO_t	12.55	12.38	11.90	12.46	12.54	7.56	13,01	11,73	13,48	12,24	12,13	11,93	12,27	12,25	12,01	14,13	12,07
MnO	0.18	0.19	0.17	0.18	0.19	0.17	0,20	0,18	0,20	0,17	0,18	0,17	0,17	0,19	0,18	0,20	0,17
MgO	4.59	3.34	5.37	5.08	4.94	2.33	3,85	4,55	3,63	4,38	3,91	4,24	4,43	3,49	3,72	3,89	4,45
CaO	8.17	6.68	8.90	7.91	8.17	4.13	6,79	5,82	7,40	7,48	7,06	7,72	8,32	5,17	6,95	7,38	8,59
Na_2O	2.76	3.32	2.58	2.53	2.53	3.68	2,72	3,52	2,87	2,59	2,90	2,77	2,67	3,25	3,13	2,65	2,82
K_2O	1.84	2.39	1.70	1.93	1.88	3.38	2,27	2,34	1,81	1,97	1,94	1,74	1,69	2,84	2,06	2,12	1,60
P_2O_5	0.79	1.02	0.58	0.73	0.74	0.69	1,42	0,81	0,60	0,88	0,71	0,76	0,70	1,30	0,90	0,68	0,59
Fe_2O_3	5.21	3.72	2.66	3.56	4.09	4.68	4,50	3,33	5,20	3,31	4,38	3,16	3,42	3,74	3,26	5,39	3,14
FeO	7.86	9.03	9.51	9.26	8.86	3.35	8,69	8,53	8,49	9,01	7,97	8,87	8,92	8,60	8,87	8,90	9,07
L.O.I.	1.91	1.87	1.50	1.84	1.90	2.87	1,98	1,66	2,41	1,94	1,72	1,71	2,13	2,22	1,61	2,59	1,36
mg number	0.43	0.35	0.48	0.45	0.44	0.38	0,37	0,44	0,35	0,42	0,39	0,42	0,42	0,37	0,39	0,36	0,43
Rock Type	AB	LA	AB	AB	AB	DAC	LA	LT	AB	AB	AB	AB	AB	LA	AB	AB	AB
Cr (ppm)	85	14	170	120	119	1	16	19	34	90	37	19	17	8	12	19	73
Ni	48	10	71	52	50	79	8	26	37	42	31	41	42	6	22	13	65
Ba	570	721	464	556	533	2280	727	632	536	572	607	507	490	823	628	581	451
Rb	39	56	32	39	37	98	43	45	34	33	33	34	28	62	46	46	25
Sr	591	609	673	537	555	705	533	462	545	553	531	570	582	482	574	598	622
La	41	49	29	39	41	62	56	51	52	50	55	39	47	71	55	53	36
Ce	97	121	69	97	94	147	107	93	91	99	108	90	77	140	112	90	69
Nd	52	64	40	47	50	83	74	63	60	63	67	53	50	86	63	57	41
Zr	304	404	236	284	282	522	344	336	323	279	300	227	211	451	314	319	187
Y	47	55	38	43	45	62	60	51	51	46	49	41	41	66	49	51	36
Nb	26	34	19	24	24	34	27	26	28	23	25	20	20	33	27	25	17

Oxides are normalized to 100% on a volatile free basis, with total iron expressed as FeO_t. Original Fe_2O_3, FeO and loss on ignition (L.O.I.) are also reported; mg number, $Mg/(Mg+Fe^{2+})$, is calculated assuming $Fe_2O_3/FeO = 0.15$. SF sites are the reanalyzed samples from the Sardinha Formation [*Bellieni et al.*, 1990]; sites 1 to 31 are indicated in Fig. 2. Rock type according to *De La Roche et al.* [1980]: TB = tholeiitic basalt, AB = andesi-basalt, LA = lati-andesite, LT = latite, DAC = dacite.

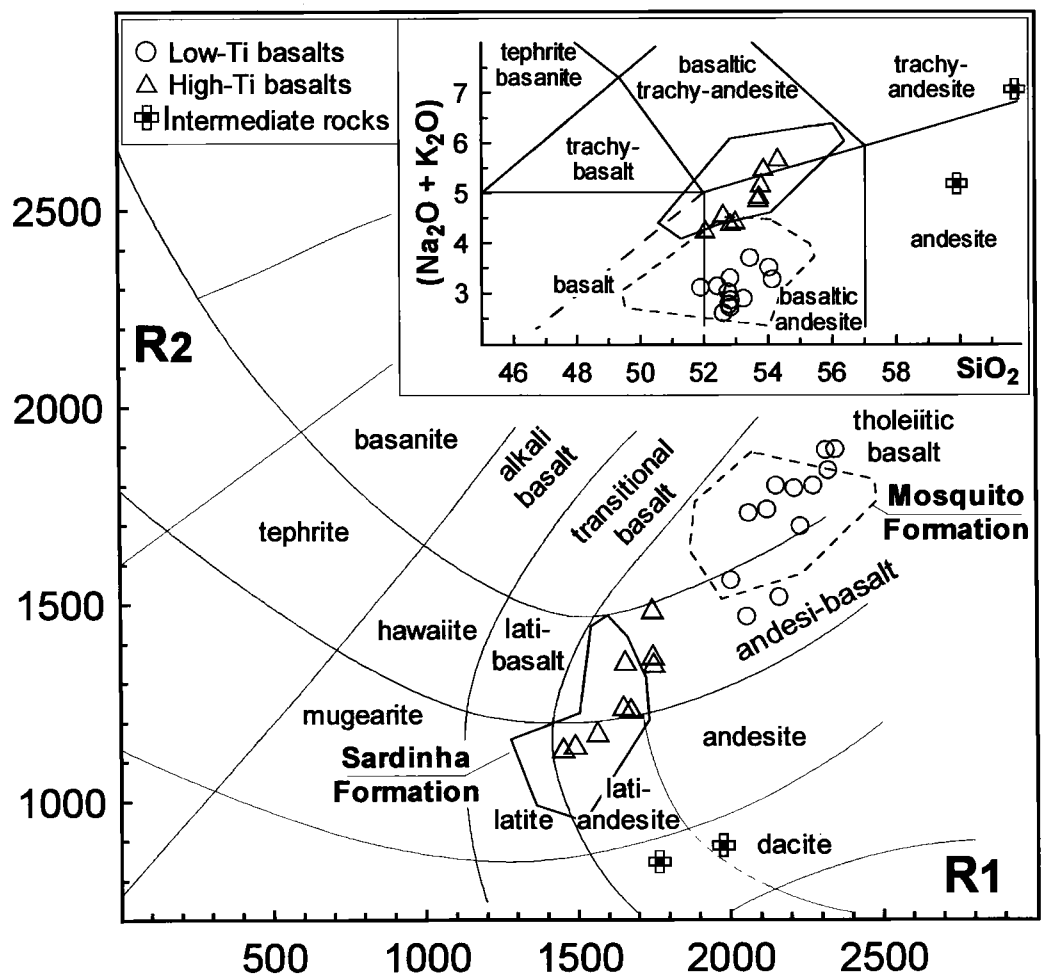


Figure 8. R1-R2 classification diagram [*De La Roche et al.*, 1980] of the low- and high-Ti rocks from the Maranhão basin and Ceará State. The fields of the Mosquito [*Bellieni et al.*, 1990; *De Min et al.*, 2000] and the Sardinha formations (*Bellieni et al.*, 1990; this paper] are also shown for comparison. The TAS classification [*Le Bas et al.*, 1986] is shown in the inset.

pole. In Fig. 11 the African Triassic pole [*Westphal et al.*, 1986] plots close to CM2, therefore strengthening this conclusion.

Based on the above evidence it comes out that the magmatic activity in northeastern Brazil started earlier than previously thought, probably during Late Triassic, extending through Early Jurassic, Early and Late Cretaceous, and ending in the Tertiary, as testified by various flows and plugs located towards the Northern border of the Rio Grande do Norte and Ceará States. This recurrent magmatic activity in the area could be responsible for a resetting of the Ar system in the older rocks without causing a reset of the magnetization in dikes II and IV. This hypothesis is supported by the indication given by the susceptibility and coercivity variation (Fig. 12) within the Ceará-Mirim dike swarm. The fact that the susceptibility in subswarm II tends to be lower than that of the Jurassic and Early Cretaceous dikes, while coercivity tends to be higher (smaller magnetic grains), is consistent with a colder crust by the time the Triassic dikes were emplaced, which were re-heated by the younger magmatic pulses. Secondary magnetizations identified in some sites of subswarms II, III, and IV (Fig. 7) give another indication for a possible alteration of the radiometric ages. Most of the sites where the secondary component A was found belong to subswarm II; this component fits well the APWP for South America at ~130 Ma (pole A in Fig. 11), indicating that this magnetic imprint may be of thermal origin, which was probably accompanied by argon losses. Secondary component B was identified mostly in samples from subswarm IV

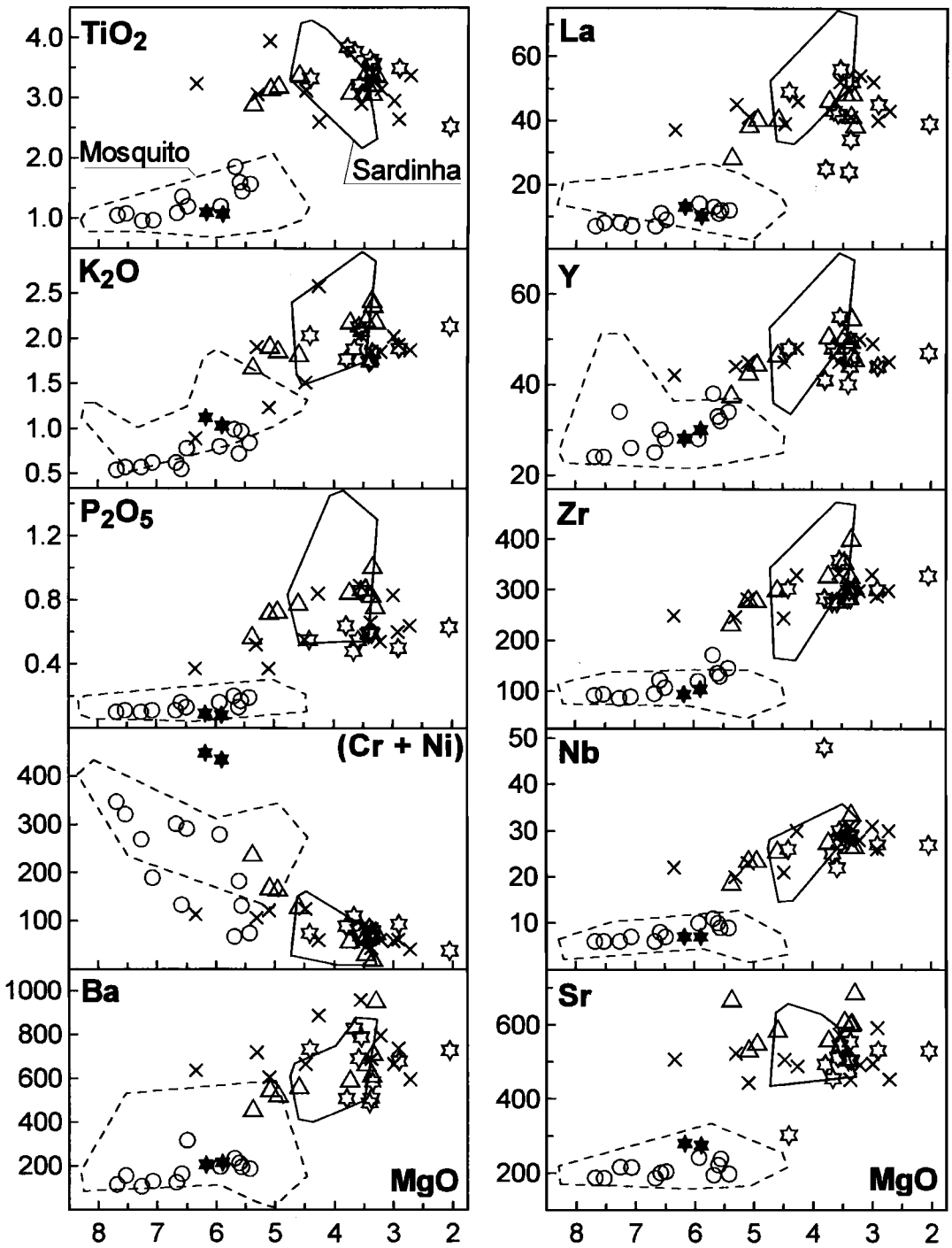


Figure 9. Variation diagrams of major, minor (wt.%), and trace elements (ppm) of the basic rocks from the Maranhão basin and Ceará State. The high-Ti rocks of the Ceará-Mirim dike swarm and the low-Ti tholeiites from the Lavras da Mangabeira, as well as the fields of the Mosquito and Sardinha formations are shown for comparison. Symbols: open circles and triangles = low- and high-Ti tholeiites from the Maranhão basin and Ceará State, respectively; solid stars = Lavras da Mangabeira flows; oblique crosses and open stars = Ceará-Mirim subswarms I, III and IV, and subswarm II, respectively.

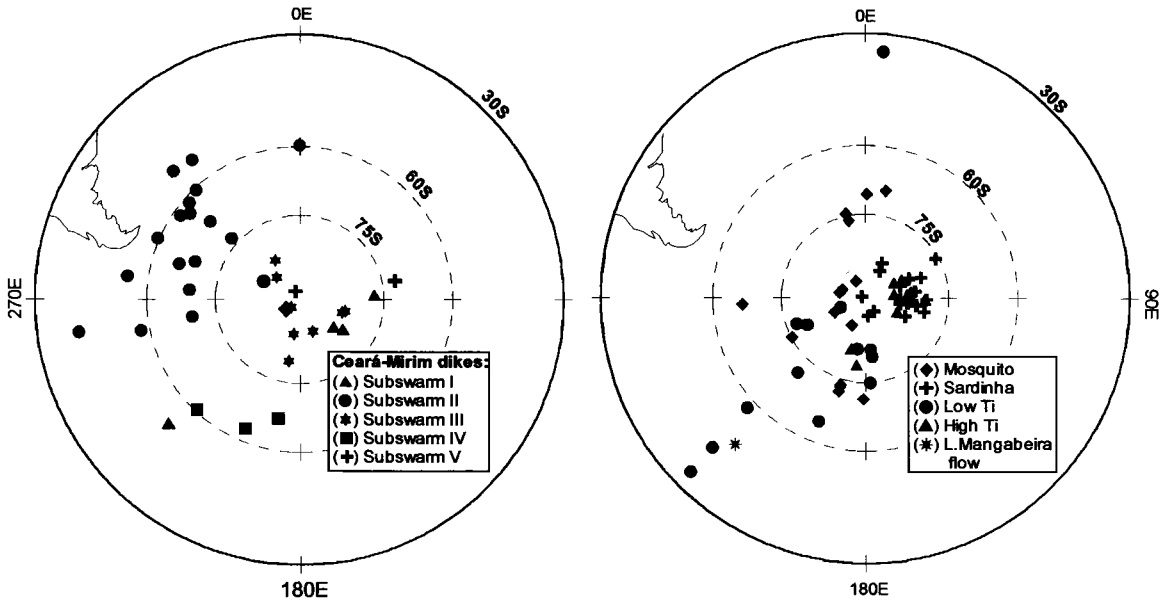


Figure 10. Virtual geomagnetic poles plotted in the southern hemisphere for a) the Ceará-Mirim dikes, and b) the Maranhão basin igneous rocks, Ceará dikes, and the Lavras da Mangabeira flow plotted in the southern hemisphere. Data from *Bücker et al.* [1986], and *Schult and Guerreiro* [1979] are also included.

(Jurassic), as well as in some from the subswarm II, and seems to be of Early Jurassic age (pole B in Fig. 11), in spite of the large uncertainty as indicated by the statistical parameters (Fig. 7).

One of the conclusions that can be addressed from the data presented in this paper is that the subdivision of age provinces in the Maranhão basin (Early Jurassic Mosquito Formation on the west, and the Early Cretaceous Sardinha Formation on the east) must be reconsidered. The northeastern part of the Maranhão basin in Piauí State is dominated by low-Ti tholeiites which present remanent magnetization and geochemical signature of Early Jurassic age, allowing to consider these rocks as correlating with the Mosquito Formation. Based on this new spatial distribution, it is possible to conclude that the Early Jurassic magmatism was more widespread and more voluminous than was supposed so far (confined to the Porto Franco-Grajaú area; Fig. 1), occurring also in the northeastern border of the basin. Eastwards, the Jurassic Ceará-Mirim subswarm IV and the low-Ti Ceará dikes testify that the CAMP-related (~200-190 Ma) activity in the extreme northeastern South America was more expressive than the limited area occupied by the remnants of the Lavras basin flows could indicate. Furthermore, it is possible to conclude that the CAMP activity in NE Brazil was preceded by Late Triassic magmatism, possibly at ca. 215-220 Ma based on the similar ages found in the magmatic activity in northwestern Africa [*Maluski et al.*, 1995]. The small sedimentary basins in northeastern Brazil, mostly filled with Mesozoic sediments, may show conglomeratic sandstones of Paleozoic age (e.g. the Araripe and Iguatú basins; *Priem et al.*, 1978], indicating that extension tectonics occurred before the emplacement of the Early Jurassic CAMP magmatism. *Hames et al.* [2000] proposed a geodynamical model for the formation of the Central Atlantic Magmatic Province (Appalachian-Mauritanian belt) where the CAMP incubating melt (ca. 230 Ma) was responsible for the subsidence of onshore basins. The subsequent (ca. 230-200 Ma) extensional tectonics and crustal thinning of the Appalachian orogen would have later (ca. 200 Ma) promoted the emplacement of the CAMP magmatic activity from Roraima plateau northwards. Similarly, in NE-Brazil extensional tectonics, and crustal thinning [*Castro et al.*, 1998] allowed the local emplacement of small volumes of Late Triassic magmatism (dikes of the Ceará-Mirim subswarm II) about 20-30 My before the huge and widespread Early Jurassic CAMP magmatic activity.

CONCLUDING REMARKS

The Mesozoic magmatism in NE Brazil is represented by two contrasting rock types, i.e. the low-Ti and high-Ti tholeiites.

In general the low-Ti tholeiites from the Northeastern Maranhão basin, Ceará and the Lavras basin show strong geochemical affinity with the Mosquito tholeiites. Petro-

Table 4. Mesozoic paleomagnetic poles for South America

Formation	Age (Ma)	N	Paleomagnetic Pole Long. (°E)	Lat. (°S)	α_{95} (°)	References
Mean Late Cretaceous	70-90*	5	349.7	80.3	4.1	Calculated from *Montes-Lauar et al.* [1995] and *Butler et al.* [1991]
Cabo de Santo Agostinho, Brazil	~100*	9	315.1	87.6	4.5	*Schult and Guerreiro* [1980]
Cañadón Asfalto Basin, Argentina	~116	16	159.0	87.0	3.8	*Geuna et al.* [2000]
Florianópolis Dikes, SE Brazil	~121†	65	3.3	89.1	2.6	*Raposo et al.* [1998]
Central Alkaline Province, Paraguay	127-130*	75	62.3	85.4	3.1	*Ernesto et al.* [1999], and reference therein
Ponta Grossa Dikes, SE Brazil	129-131†	115	58.5	84.5	2.0	*Ernesto et al.* [1999], and reference therein
Serra Geral Formation., Paraná Basin	133±1†	339	90.1	84.3	1.2	*Ernesto et al.* [1999]
Cordoba Province, Argentina	133-115*	55	75.9	86.0	3.3	*Geuna and Vizán* [1998]
Northeastern Brazil Magmatism-1	125-145	44	97.6	85.2	1.8	This paper
Northeastern Brazil Magmatism-2	175-198	33	223.9	78.1	5.2	This paper
Marifil Formation, Argentina	174-164*	8	133.4	77.7	19.0	*Vizán* [1998], and reference therein
Chon-Aike Formation, Argentina	168±2#	54	197.0	85.0	6.0	*Vizán* [1998], and reference therein
Lepá-Osta Arena Formations, Argentina	Je	13	129.4	75.5	6.8	*Vizán* [1998]
Guacamayas Group and Bolivar dikes, Venezuela	195-199*	10	265.8	76.1	11.3	*Rapalini et al.* [1993], and reference therein
French Guyana CAMP Dikes	198-192†	26	235.1	81.2	4.0	*Nomade et al.* [2000]
Guyana dykes	Tr-J	7	222.0	63.3	12.4	*Rapalini et al.* [1993], and reference therein
Caciporé Dikes, North Brazil§	~200-192†	17	208.6	79.8	5.2	*De Min et al.* [2000]; unpublished data
Roraima Dikes, North Brazil§	198.8±2.0†	7	235.1	80.1	6.6	*De Min et al.* [2000]; unpublished data
Penatecaua Dikes, North Brazil§	164±26*	9	249.5	65.6	8.9	*Guerreiro and Schult* [1986]
Anari-Tapirapuã Lavas, Brazil§	~197†	15	250.3	65.5	3.6	*Montes-Lauar et al.* [1994]
Ceará-Mirim Dike II, NE Brazil	245-160*	18	297.5	65.1	6.5	This paper
Suriname Dikes	242±12*	10	320.0	81.9	7.1	*Rapalini et al.* [1993], and reference therein
Puesto Viejo Formation, Argentina	232	6	236	-76	18.0	*Rapalini et al.* [1993], and reference therein
Ischigualasto Volcanics, Argentina	224*	9	239.2	79.2	15.0	*Rapalini et al.* [1993], and reference therein
Cacheuta Group, Argentina	Tr		266.0	74.0	14.0	*Rapalini et al.* [1993], and reference therein
C. Terneros Volcanics, Argentina	Tr		228.0	80.0	10.0	*Rapalini et al.* [1993], and reference therein
Amaná Formation, Argentina	Tr	21	317.0	83.0	8.0	*Rapalini et al.* [1993], and reference therein

N= number of sites; § Brazilian CAMP poles; *K-Ar ages; † $^{40}Ar/^{39}Ar$ ages; #Rb-Sr ages;

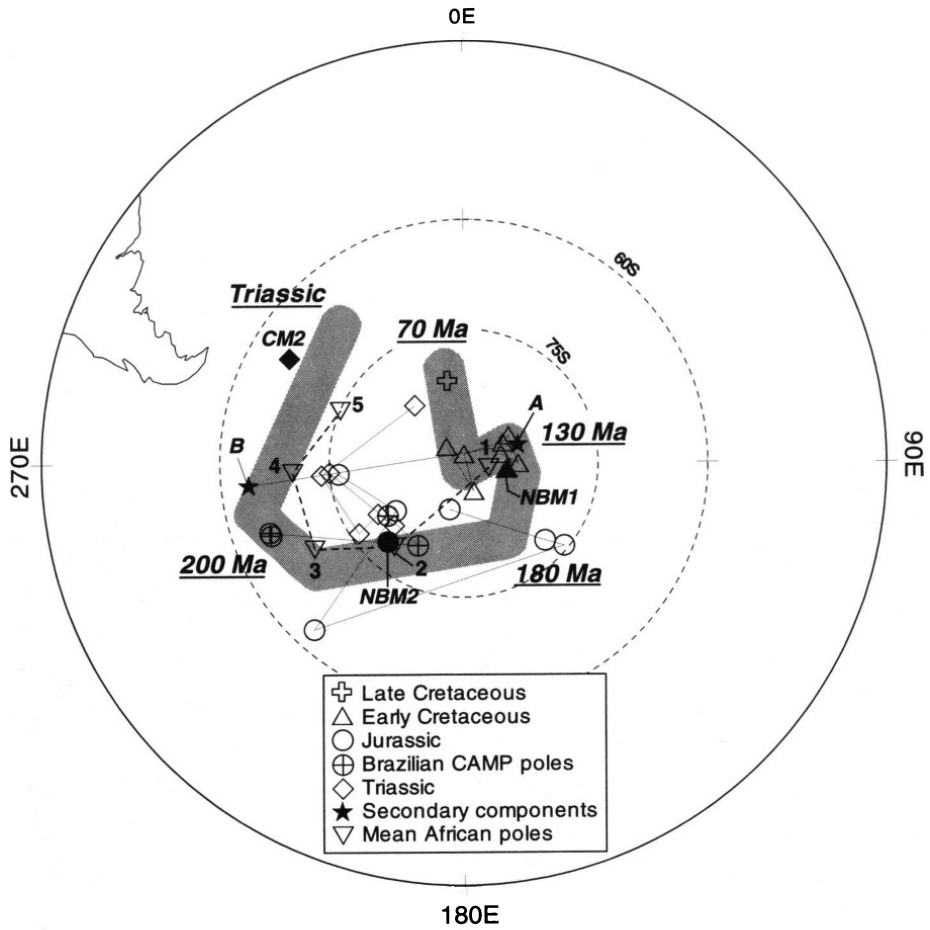


Figure 11. Mesozoic paleomagnetic poles for South America (Table 4). Full symbols represent the poles calculated in this paper. Full stars are the paleomagnetic poles corresponding to the secondary magnetization components A and B. Mean African poles rotated to South America; codes 1 to 5 refer to ages 130 Ma [*Gidskehaug et al.*, 1975], 154-180 Ma, 180-205 Ma, 205-230 Ma [*Kies et al.*, 1995], and 240-245 Ma [*Westphal et al.*, 1986], respectively.

logic similarities also exists between the dikes from the Ceará-Mirim (subswarms I and III), the Ceará high-Ti rocks, and the Sardinha intrusives.

Remanent magnetizations allow distinguishing distinct sets of Ceará-Mirim dikes as proposed by *Sial* [1975]. Remanences also permit to discriminate the low- and high-Ti groups from the Maranhão basin (Mosquito and Sardinha formations, respectively) and other intrusives from the Ceará State.

Paleomagnetic poles calculated for the Early Cretaceous (NBM1; Ceará-Mirim I, III and V, the Sardinha Formation, and other high-Ti intrusives), and the Early Jurassic (NBM2; Ceará-Mirim IV, Lavras basin flows, Mosquito Formation and other low-Ti intrusives) are in accordance with other South American and African paleomagnetic poles of same age.

A Late Triassic age is proposed for the paleomagnetic pole from the Ceará-Mirim subswarm II (CM2), which differs from the other poles of the studied area. Two K-Ar ages older than 200 Ma have already been reported for this intrusives, and the pole position agrees with coeval poles from Africa and South America.

The new paleomagnetic and geochemical data presented in this paper demonstrate that the previous age subdivision for the Mesozoic magmatic rocks in the Maranhão basin (Early Jurassic Mosquito Formation on the west, and the Early Cretaceous Sardinha Formation on the east) must be reconsidered. The northeastern part of the basin (in Piauí State) is dominated by low-Ti rocks of Early Jurassic age corresponding to the Mosquito Formation.

The Early Jurassic (CAMP) magmatism in the Maranhão basin was more widespread and voluminous than was

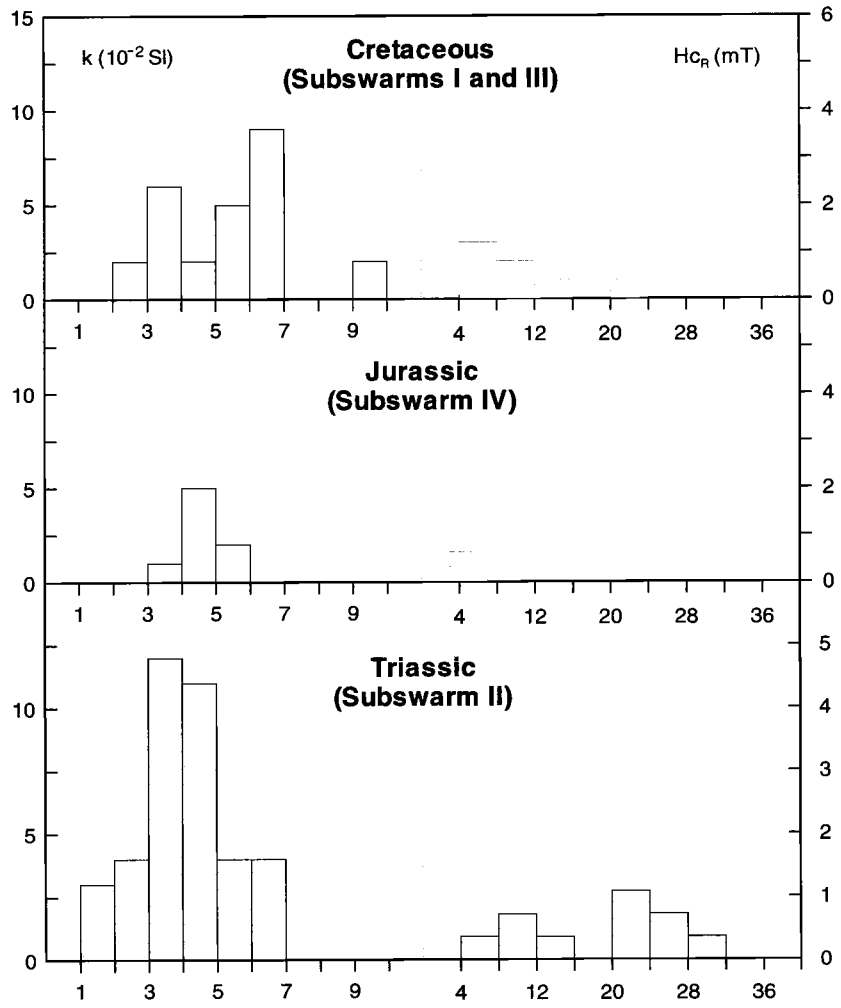


Figure 12. Histograms of susceptibilities in 10^{-2} SI units (on the left), and coercivities in mT (on the right) for the Ceará-Mirim dikes.

supposed so far, occurring also in the northeastern border of the basin. Eastwards, the Ceará-Mirim subswarm IV testifies that the CAMP-related activity in the extreme northeastern South America was more expressive than the limited area presently occupied by the remnants of the Lavras basin flows.

The Ceará-Mirim subswarm II represents a Late Triassic magmatism which preceded the CAMP activity in the Northeastern Brazil.

Acknowledgments. This project has been supported by the Brazilian agencies PADCT/FINEP (Project 0568/92), FAPESP (Project 93/2796-1) and CNPq (40.4085/91-1), and by the Italian agencies MURST and CNR. Thanks are due to J. Marins and R. Siqueira for the help in the various phases of the paleomagnetic work. The authors are grateful to N. Bardelli for his help in preparing figures and diagrams. Logistic facilities during field work were provided by University of Rio Grande do Norte and Petrobrás. Thanks are also due to W. MacDonald and A. Grunow by their thorough reviews of the manuscript.

REFERENCES

Almeida, F.F.M., Carneiro, C.D.R., Machado Jr., D.L., and Dehira, L.K., Magmatismo pós-Paleozóico no Nordeste Oriental do Brasil. *Rev. Brasil. Geoc.*, 18, 451-462, 1988.

Baksi, A.K., and Archibald, D.A., Mesozoic igneous activity in the Maranhão province, Northern Brazil: $^{40}Ar/^{39}Ar$ evidence for separate episodes of basaltic magmatism. *Earth Planet. Sci. Letters*, 151, 139-153, 1997.

Bellieni, G., Macedo, M.H.F., Petrini, R., Piccirillo, E.M., Cavazzini, G., Comin-Chiaramonti, P., Ernesto, M., Macedo, J.W.P., Martins, G., Melfi, A.J., Pacca, I.G., and De Min, A., Evidence of magmatic activity related to Middle Jurassic and

Lower Cretaceous rifting from NE-Brazil (Ceará-Mirim): K/Ar age, palaeomagnetism, petrology and Sr-Nd isotope characteristics, *Chem. Geology*, 97, 9-32, 1992.

Bellieni, G., Piccirillo, E.M., Cavazzini, G., Petrini, R., Comin-Chiaramonti, P., Nardy, A.J.R., Civetta, L., Melfi, A.J., and Zantedeschi, P., Low- and High-TiO_2 Mesozoic tholeiitic magmatism of the Maranhão basin (NE-Brazil): K/Ar age, geochemistry, petrology, isotope characteristics and relationships with Mesozoic low- and high-TiO2 flood basalts of the Paraná basin (SE-Brazil), *Neues Jhr. Miner. Abh.*, 162, 1-33, 1990.

Bellieni, G., Brotzu, P., Comin-Chiaramonti, P., Ernesto, M., Melfi, A. J., Pacca, I.G., Piccirillo, E.M., and Stolfa, D., Petrological and paleomagnetic data on the plateau basalts to rhyolite sequences of the southern Paraná Basin (Brazil). *An. Acad. Bras. Ciênc*, 55, 355-383, 1983.

Bücker, C., Schult, A., Bloch, W., and Guerreiro, S.D.C., Rock-magnetism and palaeomagnetism of an Early Cretaceous/Late Jurassic dike swarm in Rio Grande do Norte, Brazil, *J. Geophys.*, 60, 129-135, 1986.

Buddington, A.F., and Lindsley, D.H., Iron-titanium oxide minerals and synthetic equivalents, *J. Petrol.*, 5, 310-357, 1964.

Butler, R.F., Hervé, F., Munizaga, F., Beck, M.E.Jr., Burmester, R.F., and Oviedo, E.S., Paleomagnetism of the Patagonian plateau basalts, southern Chile and Argentina. *J. Geophys. Res.*, 96(B4), 6023-6034, 1991.

Castro, D.L., Medeiros, W.E., Jardim de Sá, E.F., and Moreira, J.A.M., Gravity map of part of Northeast Brazil and adjacent continental margin and its interpretation based on the hypothesis of isostasy. Brazil. *J. Geophys.* 16, 115-129, 1998.

De La Roche, H., Leterrier, P., Grandclaude, P., and Marchal, M., A classification of volcanic and plutonic rocks using R1-R2 diagram and major-element analyses. Its relationships with current nomenclature, *Chem. Geol.*, 29, 183-210, 1980.

De Min, A., Piccirillo, E.M., Marzoli, A., Bellieni, G., Renne, P.R., Ernesto, M., and Marques, L.S.. The Central Atlantic Magmatic Province (CAMP) in Brazil: Petrology, Geochemistry, $^{40}Ar/^{39}Ar$ ages, paleomagnetism and geodynamic implications, in *AGU*, this volume.

Ernesto, M., Raposo, M.I.B., Marques, L.S., Diogo, L.A., and De Min, A., Paleomagnetism and geochemistry of the magmatic rocks from Northeastern Paraná Magmatic Province: regional tectonic implications. *J. Geodynamics*, 28, 321-340, 1999.

Fisher, R.A., Dispersion on a sphere. *Proceedings of Royal Society of London*, Ser. A 217, 295-305, 1953.

Fodor, R.V., Sial, A.N., Mukasa, S.B., and McKee, E.H., Petrology, isotope characteristics and K-Ar ages of the Maranhão, northern Brazil, Mesozoic basalt province. *Contrib. Mineral. Petrol.*, 104, 555-567, 1990.

Geuna, S.E., and Vizán, H., New Early Cretaceous palaeomagnetic pole from Córdoba Province (Argentina): revision of previous studies and implications for the South American database. *Geophys. J. Int.*, 135, 1085-1100, 1998.

Gidskehaug, A., Creer, K.M., and Mitchell, J.G., Palaeomagnetism and K-Ar ages of the South-West African basalts and their bearing on the time of the initial rifting of South Atlantic Ocean. *Geophys. J.R. Astron. Soc.*, 42, 1-20, 1975.

Gomes, J.R.C., Gatto, C.M.P.P., de Souza, G.M.C., da Luz, D.S., Pires, J.L., and Teixeira, W., Geologia: mapeamento regional, Folhas SB.24/25 - Jaguaribe/Natal, *Projeto Radambrasil, Levantamento de Recursos Naturais*, 23, 29-300, 1981.

Guerreiro, S.D.C., and Schult, A., Palaeomagnetism of Jurassic tholeiitic intrusions in the Amazon Basin. *Münchner Geophysik. Mitteil.*, 1, 37-48, 1986.

Guerreiro, S.D.C., and Schult, A., Paleomagnetismo de um enxame de diques toleíticos de idade meso-cenozóica, localizados no Rio Grande do Norte, *Rev. Brasil. Geof.*, 1, 89-98, 1983.

Hames, W.E., Renne, P.R., and Ruppel, C., New evidence for geologically instantaneous emplacement of earliest Jurassic Central Atlantic magmatic province basalts on the North American margin. *Geology*, 28, 859-862, 2000.

Horn, P., Müller-Sohnius, D., and Schult, A., Potassium-Argon ages on a Mesozoic tholeiitic dike swarm in Rio Grande do Norte, Brazil, *Rev. Brasil. Geociênc.*, 18, 50-53, 1988.

Kies, B., Henry, B., Merabet, N., Derder, M.M., and Daly, L., A new Late Triassic-Liasic paleomagnetic pole from superimposed and juxtaposed magnetizations in the Saharan craton. *Geophys. J. Int.* 120, 433-444, 1995.

Kirshvink, J., The least-squares lines and planes and the analysis of paleomagnetic data. *Geophys. J.R. Astron. Soc.*, 62, 699-718, 1980.

Le Bas, M.J., Le Maitre, R.W., Streckeisen, A., and Zanettin, B., A chemical classification of volcanic rocks based on the total alkali-silica diagram. *J Petrol.*, 27, 745-750, 1986.

Maluski, H., Coulon, C., Popoff, M., and Baudin, P., $^{40}Ar/^{39}Ar$ chronology, petrology and geodynamic setting of Mesozoic to early Cenozoic magmatism from the Benue Trough, Nigeria. *J. Geol. Soc.*, 152, 311-326, 1995.

Martins, G., Caracterização petrológica e geoquímica do enxame de diques máficos Rio Ceará-Mirim. M.Sc. thesis, University of São Paulo, São Paulo, 1991.

Marzoli, A., Renne, P.R., Piccirillo, E.M., Ernesto, M., Bellieni, G., and De Min, A., Extensive 200-million-year-old continental flood basalts of the Central Atlantic Magmatic Province. *Science* 284, 616-618, 1999.

Montes-Lauar, C.R., Pacca, I.G., Melfi, A.J., and Kawashita, K., Late Cretaceous alkaline complexes, southeastern Brazil: Paleomagnetism and geochronology. *Earth Planet. Sci. Lett.*, 134, 425-440, 1995.

Montes-Lauar, C.R., Pacca, I.G., Melfi, A.J., Piccirillo, E.M., Bellieni, G., Petrini, R., and Rizzieri, R., The Anari and Tapirapuã Jurassic formations, western Brazil: paleomagnetism, geochemistry and geochronology. *Earth Planet. Sci. Lett.*, 128, 357-371, 1994.

Nascimento, D.A., Gava, A., Pire, J.L., and Teixeira, W., Folha AS-24, Fortaleza, *Projeto RADAMBRASIL, MME/SG*, 212pp., Rio de Janeiro, Brazil, 1981.

Nomade, S., Théveniaut, S., Chen, Y., Pouclet, A., and Rigollet, C., Paleomagnetic study of French Guyana Early Jurassic dolerites: hypothesis of a multistage magmatic event. *Earth Planet. Sci. Lett.*, 184, 155-168, 2000.

Novello, A., Studio petrografico-geochimico di dicchi, sill e vulcaniti del Brasile Nord Orientale (Ceará, Piaui, Rio Grande do Norte), Tesi di Laurea, University of Trieste, 1995.

Oliveira, D.C., O papel do enxame de diques Rio Ceará Mirim

na evolução tectônica do Nordeste Oriental (Brasil): implicações na formação do rifte Potiguar, M.S. thesis, Federal University of Ouro Preto, Brazil, 1992.

Popoff, M., Du Gondwana á l'Atlantique sud: les connexions du fossé de la Bénoué avec les bassins du Nord-Est brésilien jusqu'á l'ouverture du golfe de Guinée au Crétacé inférieur, *J. Afr. Earth Sci.*, 7, 409-431, 1988.

Priem, H.N.A., Boelrijk, N.A.I.M., Verschure, R.H., Hebeda, E.H., Verdurmen, E.A.T., and Bon, E.H., K-Ar dating of a basaltic layer in the sedimentary Lavras Basin, Northeastern Brazil. *Rev. Brasil. Geociênc.*, 8,262-269, 1978.

Rapalini, A.E., Abdeldayem, A.L., and Tarling, D.H., Intracontinental movements in Western Gondwanaland: a palaeomagnetic test. *Tectonophysics*, 220, 127-139, 1993.

Raposo, M.I.B., Ernesto, M., and Renne, P.R., Paleomagnetism and $^{40}Ar/^{39}Ar$ dating of the Early Cretaceous Florianópolis dike swarm (Santa Catarina Island), Southern Brazil. *Phys. Earth Planet. Int.*, 108, 275-290, 1998.

Rolff, P.A.A., O pico vulcânico do Cabugi, Rio Grande do Norte, Notas Prelim. Estudos, *DNPM*, 26 pp., Rio de Janeiro, 1965.

Rezende, W.M., Bacia do Maranhão - estudos dos processos de intrusões e extrusões de magmas básicos, 23pp., *Petrobrás/Depex*, Rio de Janeiro, 1964.

Schult A., and Guerreiro, S.D.C., Palaleomagnetism of Upper Cretaceous volcanic rocks from Cabo de Sto. Agostinho, Brazil. *Earth Planet. Sci. Lett.*, 50, 311-315, 1980.

Schult A., and Guerreiro, S.D.C., Palaeomagnetism of Mesozoic igneous rocks from the Maranhão Basin, Brazil, and the time of the opening of the South Atlantic. Earth Planet. Sci. Lett., 42, 427-436, 1979.

Sial, A.N., Long, L.E., Pessoa, D.A.R., and Kawashita, K., Potassium-argon ages and strontium isotope geochemistry of Mesozoic and Tertiary basaltic rocks, Northeastern Brazil, *An. Acad. Brasil. Ciênc.*, 53, 116-122, 1981.

Sial, A.N., The post-Paleozoic volcanism of Northeast Brazil and its tectonic significance, *An. Acad. Brasil. Ciênc.*, 48(supl.), 299-311, 1976.

Sial, A.N., Petrologia e significado tectônico dos diabásios mesozóicos do Rio Grande do Norte e Paraíba, in *Atas VII Simpósio de Geologia*, Fortaleza, 207-221, 1975.

Vizán, H., Paleomagnetism of the Lower Jurassic Lepá and Osta Arena formations, Argentine Patagonia. *J. South Am. Earth Sci.*, 11, 333-350, 1998.

Westphal, M., Bazhenov, M.L., Lauer, J.P., Pechersky, D.M., and Sibuet, J.C., Paleomagnetic implications on the evolution of the Tethys belt from the Atlantic ocean to the Pamirs since the Triassic. *Tectonophysics*, 123, 37-82, 1986.

Zijderveld, J.D.A., AC demagnetization of rocks, analysis of results, in *Methods in Paleomagnetism*, edited by D.W. Collinson, K.M. Creer and S.K. Runcorn, 254-286, Elsevier, Amsterdam, 1967.

A.de Min and E.M. Piccirillo, Dipartimento di Scienze della Terra; Universita di Trieste, Via E. Weiss 8, 34127 Trieste, Italy.

G. Bellieni, Dipartimento di Mineralogia e Petrologia, Universita di Padova; Corso Garibaldi, 37, 35137 Padova, Italy.

G. Martins, Departamento de Geologia, Universidade Federal do Ceará, Campus do Pici, 60455-760 Fortaleza, Brazil.

I.G. Pacca, L.S. Marques and M. Ernesto, Departamento de Geofisica, Universidade de São Paulo, Rua do Matão, 1226, Cidade Universitária; 05508-900 São Paulo, Brazil, marcia@iag.usp.br.

J.W.P. Macedo, DFTE/CCET, Universidade Federal do Rio Grande do Norte, 59072-970 Natal, Brazil

View publication stats